



Hybrid deep learning approaches for the detection of diabetic retinopathy using optimized wavelet based model

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ABSTRACT

Diabetic retinopathy (DR) is a diabetic ocular manifestation that leads to loss of visual impairment and blindness in people worldwide. Detecting and diagnosing the DR remains a major question in this task. This developed work uses a robust hybrid binocular Siamese with a deep learning approach to classify the DR image. Initially, a pre-processing stage is introduced to remove unwanted noises. To perform this cross guided bilateral filter (CGBF) approach is emphasized. After the pre-processing, the feature extraction stage is presented to extract the features from the processed image. A wavelet-based Chimp optimization algorithm (WBCOA) is established for the extraction of features. After feature extraction, segmentation of optical disc (OD) and blood vessel (BV) is done via open closed watershed management (OCWSM). For the classification, binocular Siamese based AlexNet and GoogleNet with the SVM model are proposed in this work. The segmented OD and BV are input to the proposed hybrid DL network. Finally, the extracted images are fused and classified using the Support Vector Machine (SVM) model. The proposed method is implemented in Python and tested on DIARETDB0 (DB0) and DIARETDB1 (DB1) datasets. The proposed hybrid DL network attained 94% and 94.83% accuracy on DB0 and DB1 datasets, respectively. Also, the proposed model's outcomes are compared with various existing approaches. The proposed method conducted statistical analysis for the DR image based on mean and standard deviation (SD) to obtain an efficient output. These outcomes prove that the proposed hybrid DL system is accurate for early DR detection and deliver effective treatment of diabetes.

1. Introduction

Diabetic Retinopathy (DR) is one of the serious illnesses in the eye that causes the loss of vision and causes blindness in people [1]. Diabetes Mellitus is often called diabetes [2]. About 80 % of the people have been affected by DR for over 10 years. There are three types of diabetes: type 1 diabetes [3], type 2 diabetes [4], and gestational diabetes [5]. Among these diabetes types, 2 diabetes is the most commonly affected. DR is occurred by impairing the blood vessels in the tissues and retina [6]. The main risk factor is low blood sugar levels. A person with long-term diabetes has more chance of occurring DR [7]. Life expectations and quality are contracted due to micro-and macro cardiovascular problems [8]. So, the detection and classification of diabetes are significant. Early detection and classification [9] assist in suppressing people from loss of vision or minimizing the progress of DR. Blood vessel leakage, abnormal growth of new blood vessels, nerve tissue damage, retinal swelling and

aneurysms [10] are the various symptoms of DR.

The DR is detected in stages like moderate, severe and mild [11]. The detection of DR is challenging due to miscommunication, delayed results, treatment, etc. A specialist can detect DR based on the abnormal changes connected with vascular abnormalities [12] developed by the disease. Due to this, screening a person [13] who has diabetes is advisable. In the starting stage, the impairments caused in the retina [14] are detected by retinal screening. DR is symptomless at the starting stage, so many people are unaware of the condition. Manual grading requires better expertise and experience. Immediate DR detection is required in some places with high rates of diabetes. The development of lesions in the fundus images [15] assists in detecting the DR at the starting stage. Nowadays, many efforts are taken to provide computerized scanning systems using color funded images. However, the detection of DR is still considered an open issue, and more improvements are required to establish an alternative solution. Besides, developing an

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intelligent diagnosis system is considered along with a computer vision based system [16] and fundus images.

In the modern world, Artificial Intelligence and machine learning based approaches are more commonly used to improve outcomes and minimize efforts. Various problems are solved using intelligent approaches. The detection of diabetes using an intelligent system will minimize the efforts of humans [17]. The doctors detect diabetes based on the symptoms the same way the machine learning approaches [18] use some parameters to detect the person. The machine learning systems are trained to identify whether the person is diabetic or not. Many applications are used to detect that deep learning is the current approach [19] used to detect the person. Image processing is a significant method [20] used to attain medical image data.

In recent years, many researchers have been fascinated by the DL approach and established the best outcome in classifying DR images. Earlier, only a few studies were undertaken for the detection and diagnosis the DR affected patients. In order to improve the vision, a new approach must be enforced to obtain better accuracy. Previous approaches have suffered greatly due to large disadvantages in detecting DR. The major limitations include lack of proper operation, noisy outcome, high computational cost, and slow detection of DR due to the lack of modern techniques. These major drawbacks are the major motivation to develop the hybridized DL approach in this work to overcome the limitations above. In the proposed work, a hybridized AlexNet-GoogleNet-SVM model is introduced to detect and diagnose DR affected patients efficiently. This model obtains fast operation in training and testing the datasets based on hierarchical features. The CGBF approach is introduced in the pre-processing stage to filter out the noise. The pre-processed image is then given to extract the features based on wavelet based chimp optimization algorithm. The extracted images are then segmented using the open-closed watershed management (OCWSM) technique to segment the BV and OD. Finally, the extracted and segmented images are given to the concatenate layer to fuse the features. The SVM classifier is used to classify the outcome as normal and abnormal. The major contributions of the proposed method are clearly explained below:

- This research mainly aims to detect the presence and absence of DR using a hybrid DL (binocular Siamese based AlexNet and GoogleNet with SVM model) approach. The proposed technique helps to detect and classify the accurate presence and absence of DR in the retinal image.
- The pre-processing stage is introduced for the deletion of noise and to improve the image quality, and the CGBF technique does it. The image re-sizing and data augmentation operation also occur in the pre-processing stage.
- The feature extraction stage is emphasized using the WBCOA technique to improve the detection accuracy and preserve the information. This wavelet based optimization strategy is emphasized to extract some essential features like Shannon entropy, Kapur entropy, renyi entropy and texture features.
- To simplify the detection process, segmentation is contemplated in this work and done using the OCWSM technique. The OD and BV are segmented from the retinal image.
- The hybrid DL network allows the extraction of the most discriminative features and eliminates redundant information. Then, the feature is fused in the concatenate layer and classified using the SVM model.
- The performance is analyzed using accuracy, precision, F-measure, specificity, recall, FNR, NPV, FDR, and MCC for DBO and DBI datasets, respectively.

The following sections are structured below: Section 2 explains the review of recently issued papers on DR detection. Section 3 emphasizes the proposed methodology. Section 4 evaluates the results and discussion. Section 5 explains the conclusion of the work.

2. Literature review

As the DR rate increased, various methods were established to detect DR. K. Shankar et al. [21] proposed a deep neural network with a moth search optimization (DNN-MOS) algorithm for DR image detection and classification. It contains various methods such as pre-processing, segmentation, feature extraction and classification. First, a Contrast limited adaptive histogram equalization technique was used to enhance the contrast level of the DR image. The histogram model was introduced to segment the DR image after pre-processing. Next, the feature vectors were extracted, and the extracted features were classified using the DNN-MDO classifier process.

In the early stages of DR, there are no symptoms, leading to blindness in the person affected by diabetes. But diagnosing at the early stage minimizes the effect of DR. Javeria Amin et al. [22] proposed an automated method for diagnosing and classifying DR. This method integrates pre-processing, segmentation, feature extraction and classification. The region of interest was improved by applying a local contrast enhancement method in the grey scale image. The accurate lesions region was segmented using an adaptive threshold model and mathematical morphology. Then, statistical and geometric features were combined for improved classification. As a result, better accuracy was attained.

The persons affected by DR could bear haemorrhages, clots, lesions etc., in the retinal region. The retinal fundus photographs diagnose the situation of DR. Sanskruti Patel [23] proposed a DR detection and classification method based on a pre-trained Convolutional Neural network (CNN) to identify the patterns. Initially, a pre-processing and augmentation method was applied to enhance the extracted features. The noise present in the image was minimized by applying Gaussian blur to the fundus image. In the experimental scenario, accuracy was calculated.

Shraddha Das et al. [24] proposed a deep learning technique for extracting features from the segmented fundus image and classification of DR. First, the fundus image was pre-processed, and then the pre-processed image was segmented. The maximal principal curvature model was introduced to take out the branched blood vessels, and it uses the highest Eigenvalues. A morphological opening and an adaptive histogram equalization technique were applied to enhance and ignore the false area. A higher optic nerve proliferation has been found in diabetics. The DR classification was done using CNN. The CNN contains the bottleneck layer, convolution layer and pooling layer. In an experimental scenario, the precision of 97.2 % and accuracy of 98.7 % were obtained. However, the suffering period of the patients could not be identified.

Laxmi Math & Ruksar Fatima [25] proposed a DR classification model using an Adaptive machine learning approach. This approach supports the image's classifiers and features and identifies the DR images. The segment level Diabetic retinopathy estimation (DRE) was obtained using adapted pre-trained CNN. Then, the DR image classification was determined by connecting all segmented level Diabetic retinopathy maps (DRM). Further, the non-uniform lesions were handled end-to-end based on a learning strategy. Finally, 96.37 % of specificity and 96.37 % sensitivity were determined. Moreover, this method could not consider proliferative diabetic retinopathy.

Because the visual view of the retinal images was affected during the screening of the DR by the ophthalmologists to identify the symptoms like microaneurysm, haemorrhage, etc., this process is complicated and wastes time. So Abdul Hafiz Abu Samah et al. [26] proposed an automated machine learning algorithm to detect DR from fundus images. The pre-processing enhancement approach was used to improve the accuracy of classifying the DR images using deep learning. The vision standard of the fundus image was enhanced by using a de-haze algorithm, histogram equalization, and high pass filter during image enhancement. For image classification, four-layered convolution was involved. At last, a moderate accuracy was attained as an outcome.

Lu et al. [27] introduced the TBNNet: a context-aware graph network

for tuberculosis diagnosis. The conventional CNN model was highly suffered due to training incapability and overfitting problems. Thus, transfer learning based techniques were introduced in this work to extract some of the sample level features. In addition, efficientNet was emphasized as a pre-trained model for training the number of images in the dataset. The spatial relation between feature vectors was very much useful for classification purposes. These feature vectors can provide similar characteristics to the adjacent feature vectors. Here, features were preserved by providing a feature graph for preserving the spatial features of the image. In Lu et al. [28], the author introduced the improved AlexNet and ELM with a chaotic bat algorithm. Here AlexNet based transfer learning approach was introduced by training the image datasets. The parameter training was hard and time consuming. In this technique, batch normalization (BN) was introduced to enhance the robustness of the AlexNet model. In addition, the ensemble learning model (ELM) was introduced to replace several layers in the AlexNet model. Thus, the accuracy of the model gets increased.

Jadhav et al. [29] introduced the analysis of DR detection by segmenting the BV and OD and retinal abnormalities. This method has three major stages: pre-processing, segmentation, feature extraction and classification. In the pre-processing stage, the RGB scale was converted into a green channel component using contrast limited adaptive histogram equalization (CLAHE) technique. After pre-processing, OD and BV were segmented using the OCWSM technique. For segmenting the abnormalities, top hat transformation and Gabor filtering were introduced. After segmentation, some features like local binary pattern (LBP), texture energy measurement (TEM) and entropy were extracted. In addition, some of the optimal features were selected using trial based bypass with an improved dragonfly algorithm (TB-DA) technique. The DBN + NN technique was introduced to classify mild, severe and moderate for the classification. In the experimental section, accuracy, precision, MCC, FNR, FPR, NPV, FDR and F1-score were analyzed and compared with existing techniques.

2.1. Problem formulation

Conducting various searches in the related work, the previous method is highly affected due to major drawbacks that result in low accuracy and lack of efficient classification outcome. In [21], the DNN-MOS algorithm for detecting and classifying DR images is manifested. But, the disease could not be detected remotely due to high computational cost. [22] introduces an automated method for diagnosing and classifying DR. However, statistical and geometric features are lacking to effectively classify the results. As a result, misclassification occurs with low accuracy. In [23], the DR detection and classification method based on pre-trained CNN to identify the patterns is introduced by a researcher. But, this approach takes high time to train the datasets due to a lack of advanced approaches. In [24], the DL technique for extracting features from the segmented fundus image and the classification of DR is emphasized. However, the suffering period of the patients could not be identified due to the lack of modern techniques. In [25], the DR classification model uses the Adaptive ML approach. But, this method could not consider the proliferative DR in the suffering patients. In [26], an automated ML algorithm detects DR from the fundus image. But, the enhanced method did not satisfy the requirements due to a minor mismatch in the outcome.

Furthermore, only a few approaches are developed to detect and diagnose DR efficiently. The above technique shows better classification results and high computational time and complexity during training the datasets. The proposed research introduced the hybrid-based deep learning approach for the immediate requirement of an advanced approach. The proposed model eliminates the error and attains better accuracy efficiently. Along with the performance metrics, statistical analyses are also provided to understand the working of the proposed model.

3. Proposed diabetic retinopathy detection model

Diabetic retinopathy is a diabetic impediment which damages the human eyes. It happens due to damage in the blood vessel tissue at the back of the retina. This article proposes a novel hybridized DL approach for DR detection from the retinal fundus image. This research undergoes four major stages: pre-processing, feature extraction, segmentation and classification. Initially, the images are pre-processed using the CGBF technique. Some significant operations are undertaken in de-noising, image re-sizing and data augmentation. The pre-processed image is then fed to the feature extraction stage for the extraction of important features from the retinal image. Using the WBCOA approach, features like Kapur, renyi, Shannon entropies and texture features are extracted. After feature extraction, the segmentation process is exhibited. Here, the OCWSM technique is introduced to segment OD and BV from the retinal image. After segmentation, binocular Siamese-based AlexNet and GoogleNet with SVM classifiers are proposed to classify normal and abnormal images. The segmented OD and BV are input to AlexNet and GoogleNet, respectively. The segmented and extracted features are then fused in the concatenation layer. Finally, the SVM classifier is considered to classify the output image instead of the softmax classifier.

Fig. 1 illustrates the workflow of the proposed Hybrid DL model. The binocular network based Siamese entity can utilize the fundus image concurrently and assemble its features to predict each eye. In addition, AlexNet and GoogleNet are employed to provide optical disc and blood vessels as input to obtain classification results. The AlexNet and GoogleNet are part of the convolutional neural network that hybrids to obtain better efficiency. The AlexNet is the pre-trained model that has eight layers with learning parameters. This model contains five max-pooling layers, 3 fully connected layers and (rectified linear activation unit) ReLu activation unit with no output layer. Finally, GoogleNet is the CNN with 22 layers that can train more than millions of images without fail. A support vector machine (SVM) can be utilized to overcome classification and regression problems. Furthermore, a data augmentation strategy is followed to handle the drawbacks based on imbalanced classes.

3.1. Pre-processing stage

For the detection of DR images, ordinary images cannot be processed. There is a need for image modifications such as re-sizing and image augmentation strategy for obtaining an efficient classified outcome. Initially, the input image is re-sized to a permanent size of 256×256 pixels. After re-sizing, the image augmentation approach is manifested. In this stage, assorting, packing and horizontal turnover are generally used to train the large neural networks (NN). In addition, the RGB image is converted to a grayscale image before pre-processing. After modifying input images, the pre-processed stage is undergone. In the proposed work, the pre-processing stage occurs via cross guided bilateral filter (CGBF) to remove unwanted noise from the image.

CGBF is one of the filter approaches with a robust property of a bilateral filter with a guided image. The guided and input images are considered for calibrating the image weights. This bilateral filter removes the image illumination and analyzes the weak, noisy signals. The input image is filtered, and the guided image is evaluated to sharpen the kernels; hence flexible edges with smoothened images are obtained. The CGBF filter with spatial and kernel are integrated to clear the additive noise. After the filtering is done, the image illumination is highly reduced. The output of the filtered image is mathematically expressed as,

$$\begin{aligned} V_{CGBF}(u, v) &= G_{CGBF} \\ &= \frac{1}{w} \sum_{U, V \in U, V} c_{np}(|(U) - (V)|) n_{pd}(|B(U) \\ &\quad - B(V)|) B(V) + f_{U,V} K_{U,V} + S_{U,V} \end{aligned} \quad (1)$$

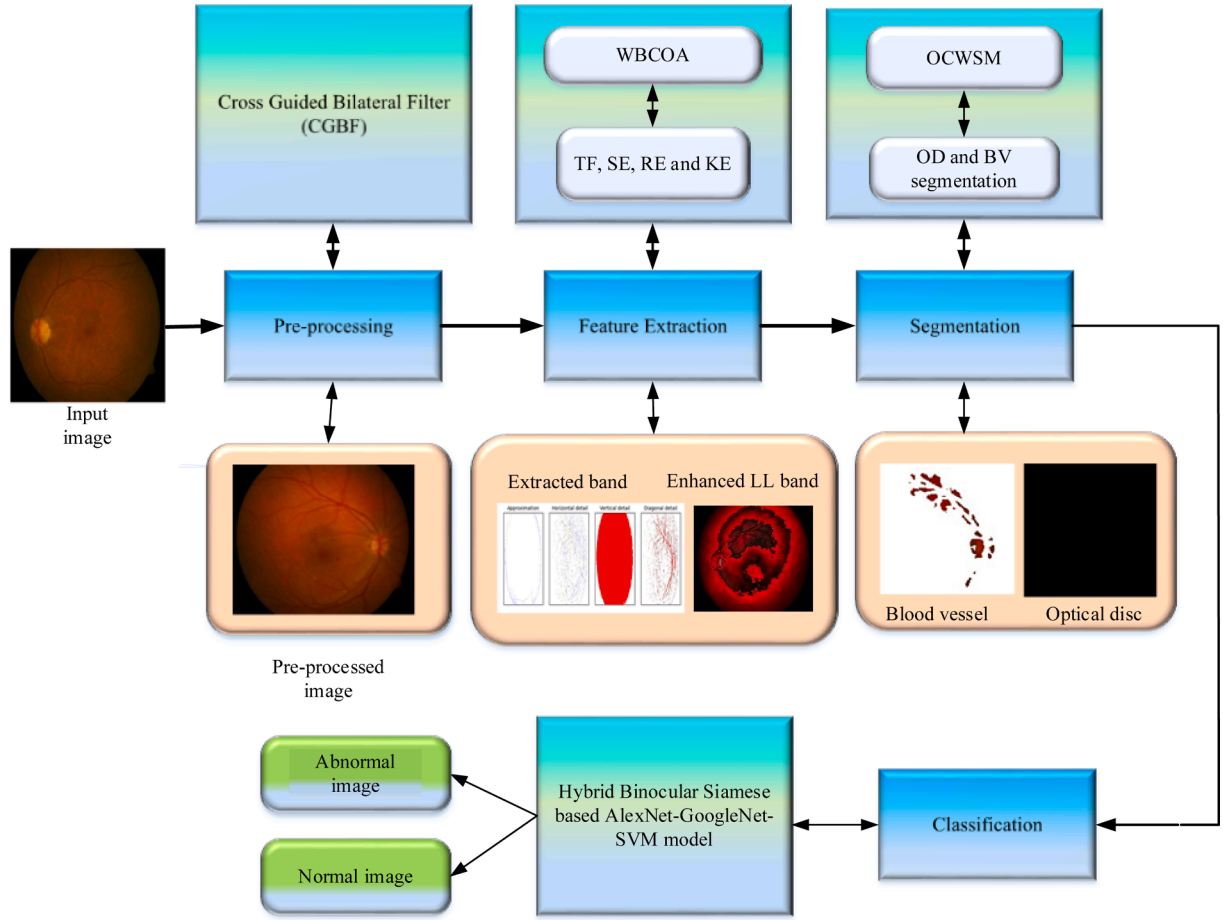


Fig. 1. Workflow of proposed Hybrid DL model.

The equation above (1) $|U - V|$ manipulates the distance among U and V pixels, where U and V are two neighbouring pixels, $B(U)$ and $B(V)$ are the location of pixels. Here, c is the guided image density.

$$n_{\phi P}|U - V| = e^{-\frac{(|U - V|)^2}{2\phi P^2}} \quad (2)$$

$$n_{\phi D}(|B(U) - B(V)|) = e^{-\frac{(|B(U) - B(V)|)^2}{2\phi D^2}} \quad (3)$$

ϕP and ϕD indicates the limitations, it dominates the bilateral filtering process. Where $n_{\phi P}$ and $n_{\phi D}$ indicates the spatial and Gaussian function. Stimulating ϕP denotes the smoothing of features that eliminate the mean square error (MSE).

$$f_{U,V} = \frac{1 \sum_{j \in U,V} K_x^j J_y^j - K_x^{U,V} J_y^j}{\phi_{U,V}^2 + E} \quad (4)$$

$$S_{U,V} = J_y^j - f_{U,V} K_x^{U,V} \quad (5)$$

$f_{U,V}$ and $S_{U,V}$ manifests the coefficients of the guided filter, and an average of the input and guided image are $K_x^{U,V}$ and $f_{U,V}$ also w evaluates the window function. The variance of the window $\phi_{U,V}^2$ and E is the systematic constant. The mathematical expression for the cost function is represented in equation (6).

$$X(f_{U,V}, S_{U,V}) = \sum_{j \in wO} (f_{U,V} K_x^j + S_{U,V} J_y^j)^2 + A f_{U,V}^2 \quad (6)$$

3.2. Feature extraction stage

After pre-processing, features are extracted from the fundus images to understand the nature of the image. For extraction of features, the proposed work utilizes hybrid optimized DWT. Some features such as optic disc, blood vessel, Shannon entropy, Kapur entropy, Rensis entropy, texture features, etc., are extracted from the retinal fundus image. Using the 2D-DWT evaluates the sub-band programming of an image based on spectral frequency components with the aid of recursive and iterative techniques. The image is manipulated in the form of LH, HL, HH and LL, encrypting the details of each image in three locations that generate the approximation in it.

The generated image details can be decayed repeatedly to manipulate the next-level information and the approximation of the images. The process is repeated for better analysis because of the doubled iteration in the images. The performance of the 2D-DWT comes from 1D-DWT, considered the initial version of the discrete wavelet transform. The mathematical formulation of the one-dimensional continuous WT of the signal $H_\omega(c, d)$ is given below,

$$H_\omega(c, d) = \int_{-\infty}^{\infty} y(m) \omega_{c,d}(m) dm \quad (7)$$

Here,

$$\omega_{c,d}(m) = \frac{1}{\sqrt{c}} \omega\left(\frac{m-c}{d}\right) \quad (8)$$

Here, $\omega_{c,d}(m)$ denotes the wavelet function, c and d are the calibrated value and the transformed values, respectively. In the practical section, the correlation between the signal and different wavelets is obtained by decaying the DWT into filtering operation with high-band and low band

filters. The 2D-DWT notation for a low band and high band filter can be expressed mathematically as,

$$y_{low}(m) = \sum_{s=-\infty}^{+\infty} y[s] H[2m - s] \quad (9)$$

$$y_{high}(m) = \sum_{s=-\infty}^{+\infty} y[s] G[2m - s] \quad (10)$$

Here, values $y_{low}(m)$ and $y_{high}(m)$ manipulates the approximation and element information generated by the low and high band filter, respectively. The downsampling (DS) evaluates the DWT by a factor of 2. The LL sub-band is considered the background component of the image. Instead, LH, HL, and HH sub-band are considered the image's expanded components. The resultant background of the image is decayed repeatedly to get next-level information and background of the image. The LL band does not obtain an optimized output except for all other bands. Hence, the chimp optimization algorithm is given to low sub-bands to achieve efficient features.

The chimp optimization algorithm is a nature inspired swarm intelligence algorithm. Most algorithms involve the exploration and exploitation stage based on iterations. The best solution is obtained using search agent circles in the exploitation stage. These search agents have enhanced capability, and some mathematical expressions govern them. ChOA is the group behavior of chimps for hunting prey. The chimps undergo four stages: driver, barrier, chaser, and attacker. The driver is used to track the movement of the prey, and the attacker introduces an attack based on anticipation of prey movement. The mathematical expression for driving and chasing is given below,

$$f = |s \cdot Y_{prey}(i) - r \cdot Y_{chimp}(i)| \quad (11)$$

$$Y_{chimp}(i + 1) = Y_{prey}(i) - g \cdot h \quad (12)$$

The chimp's behavior in the exploitation stage is given in mathematical formulation as follows,

$$h_{attacker} = |s_1 \cdot Y_{attacker} - r_1 \cdot Y| \quad (13)$$

$$h_{barrier} = |s_2 \cdot Y_{barrier} - r_2 \cdot Y| \quad (14)$$

$$h_{chaser} = |s_3 \cdot Y_{chaser} - r_3 \cdot Y| \quad (15)$$

$$h_{driver} = |s_4 \cdot Y_{driver} - r_4 \cdot Y| \quad (16)$$

$$Y_1 = Y_{attacker} - h_{attacker} \cdot g_1 \quad (17)$$

$$Y_2 = Y_{barrier} - h_{barrier} \cdot g_2 \quad (18)$$

$$Y_3 = Y_{chaser} - h_{chaser} \cdot g_3 \quad (19)$$

$$Y_4 = Y_{driver} - h_{driver} \cdot g_4 \quad (20)$$

$$Y(i + 1) = \frac{Y_1 + Y_2 + Y_3 + Y_4}{4} \quad (21)$$

$$Y_{chimp}(i + 1) = \begin{cases} Y_{prey}(h) - g \cdot h & \text{if } \phi > 0.5 \\ \text{Chaotic_value} & \text{if } \phi > 0.5 \end{cases} \quad (22)$$

The equation, as mentioned earlier, f is the random vector notation limits from 0 to 2 and s indicates the random variable limits from $-2h$ to $+2h$. Considering the chaotic map, the beginning parameter is set to 0.7 for the thresholding of the optimization problem.

The traditional ChOA algorithm consists of six chaotic maps and uses different sets of equations to upgrade the random approaches. The random approaches emphasized distinguishing the coefficients s_1, s_2, s_3 and s_4 .

In simple words, the random approach was initialized, and a single chaos map was utilized. The specified version of ChOA is denoted as

ChOA12. In other words, the chaos map is also known as gauss or mouse. The chaos map is mathematically expressed as,

$$Y_{j+1} = \begin{cases} 1 & Y_j \\ \frac{1}{\text{mod}(y, 1)} & \text{otherwise} \end{cases} \quad (23)$$

3.2.1. Extraction of entropy features:

Entropy (G) is the calibration of undesirable images manipulated with the random approach. The proposed work undergoes the extraction of Shannon entropy, Renyi entropy and Kapur entropy. However, $D(u, v)$ is considered a normal or abnormal image that has the number N_j where $j = 0, 1, 2, 3, 4, \dots, k-1$ indicates the gray scale image. The histogram of the integrated image is manifested as,

$$G_j = \frac{D_j}{A \times B} \quad (24)$$

The Shannon entropy (I) is defined as the information collected from each location is inversely proportional to the probability of occurrence. The mathematical formulation of the Shannon entropy is evaluated as,

$$I = \sum_{j=0}^{f-1} G_j \log_2(G_j) \quad (25)$$

Corresponding to this, Renyi and Kapur's entropy are also extracted. The abovementioned entropy has a higher limit than the Shannon entropy based on scattering approaches. These entropies help calculate the scatter density and regularity.

The mathematical expression for the Renyi entropy (X) is determined as follows,

$$X = \frac{1}{1 - \sigma} \log_2 \sum_{j=0}^{f-1} G_j^\sigma \quad (26)$$

Here, $\sigma \neq 0, \sigma > 0$. For the extraction of features $\sigma = 3$ is considered the limitation.

The mathematical formulation of Kapur entropy is given below,

$$P_{\sigma, \eta} = \frac{1}{\eta - \sigma} \log_2 \frac{\sum_{j=0}^{f-1} G_j^\sigma}{\sum_{j=0}^{f-1} G_j^\eta} \quad (27)$$

Here, $\sigma \neq \eta, \sigma > 0, \eta > 0$. For extracting the features, $\sigma = 0.5$ and $\eta = 0.7$ is considered the limitation.

3.2.2. Extraction of texture features:

The fundus image of pixel size 256×256 is manipulated in matrix form as $G \times H \times 3$ for each basic color, such as red, green and blue. Based on a normal and abnormal DR image, multiple granular structures are produced, which are said to be texture. Texture extraction has the property of smoothening, roughness and regular structure. The local binary pattern (LBP) and laws of texture energy based designators are used in texture feature extraction, which is explained below.

The LBP is considered an efficient texture designator and is used in a wide range of applications in texture segmentation. The LBP consists of operators that present the image by introducing the neighbourhood of each image as the middle value by utilizing the outcome of a two-class number. When all pixels are presented with LBP programming, the histogram is generated and used as the texture designator. The histogram of the textured image is mathematically expressed as,

$$G_j = \sum_{u, v} A(C(u, v) = j), \quad j = 0, \dots, d - 1 \quad (28)$$

Here, x denotes the number presentation of the LBP operator, if $Y(U) = 1$ when U is true, if $Y(U) = 0$ when U is false.

To develop the rotation invariant uniform pattern based on ring quantization, the regular formation denoted by K is obtained while integrating $LBP_{M, N}$ based on M -bit two-class number ($H_{m-1}, H_{m-2}, \dots, H_1$,

H_0).

$$K(LBP_{M,N}) = |H_{m-1} - H_0| + \sum_{p=1}^{M-1} |H_M - H_{M-1}| \quad (29)$$

The rotationally invariant regular pattern with a K-parameter of at least 2 is mathematically formulated as,

$$LBP_{M,N}^{cir2} = \begin{cases} \sum_{M=0}^{M-1} q(n_m - n_f) & \text{if } M(LBP_{M,N}) \leq 2 \\ M + 1 & \text{otherwise} \end{cases} \quad (30)$$

Multiple images can be analyzed using LBP operators by selecting the ring radius at the middle of the pixel and helping in building LBP coded images in each mode. The energy and entropy of the LBP encoded images are developed for multiple operators and used as texture designators.

In-laws texture energy, the law mask is used as a texture designator and used in multiple applications in the medical field. Using the frequency domain, the law mask uses its knowledge to extract the features from the image. The mask enhances one-dimensional notations based on five features: level, edge, spot, ripple, and wave.

The vertical one-dimensional notations are convoluted to form twenty-five 2D-filters with a size of 5×5 are given in matrix form as follows,

$$\begin{pmatrix} 1 \\ 4 \\ 6 \\ 4 \\ 1 \end{pmatrix} y(-1 \ -2 \ 0 \ 2 \ 1) = \begin{pmatrix} -1 & -2 & 0 & 2 & 1 \\ -4 & -8 & 0 & 8 & 4 \\ -6 & -12 & 0 & 12 & 6 \\ -4 & -8 & 0 & 8 & 4 \\ -1 & -2 & 0 & 2 & 1 \end{pmatrix} \quad (31)$$

$L5 \qquad E5 \qquad L5E5$

In law texture energy, the features are extracted by convoluting the 2D-mask. The texture image using $L5E5$ is manifested as follows,

$$JI_{L5E5} = I(j, k) \otimes L5E5 \quad (32)$$

According to this designator, the 2D mask excluding $L5L5$ has a negligible average. The texture image is used to integrate the brightness of all texture images and makes the designators' contrast independent. The integrated texture image is mathematically expressed below,

$$Integrate(JI) = \frac{JI_{(j,k)}}{JI_{(j,k)}^{L5L5}} \quad (33)$$

The mean of absolute values helps to obtain 5×5 masks on convoluting 25 different texture designators, and it is given by,

$$TEM_{(j,k)} = \sum_{x=-7}^7 \sum_{y=-7}^7 |JI_{(j+x, k+y)}| \quad (34)$$

Moreover, coordinating twenty-five law mask designators emphasize fourteen rotation invariant texture energy calibration. It can be formulated as,

$$JR_{E5L5} = \frac{TEM_{E5L5} + TEM_{L5E5}}{2} \quad (35)$$

The images are investigated based on 14 masks; hence, their energy is constructed and used as feature designators.

3.3. Segmentation of optical disc and blood vessel

The optical disc can be manipulated using an open-close watershed transformation. This transformation is used to overcome the dark and illuminated lesions in the DR image. The mathematical expression of the watershed-based optical disc segmentation is given by,

$$W_{sh}(g) = sp(g) \cap [\cup_x (bc(rgm_s))]^o \quad (36)$$

In the equation above, the support function is given by, $sp(g)$ and the catchment basin for regional minima (rgm_s) is evaluated by, $bc(rgm_s)$. The morphological recreation happens when the intensity of the image is changed and done using watershed elastic lines. The optic disc segmentation UE^{optic*} output is given to segment the blood vessel.

The blood vessels can be segmented using the image WR^{optic} , which can be performed using grey level thresholding. This work is based on a reference image in a bimodal histogram manner. The image's background object discriminates with the threshold hl and segments based on bimodal manipulations. The outcome of the threshold image is denoted as a binary image, and it is indicated as follows,

$$DE^{vessel} = \begin{cases} 1 & \text{if } UE^{optic*}(j, k) > jl \\ 0 & \text{if } UE^{optic*}(j, k) \leq jl \end{cases} \quad (37)$$

The image with a strength value of 1 manifests the object, and 0 indicates the background. The segmented blood vessel is indicated as, UE^{vessel} and the blood vessel illumination image $UE^{vessel*}$ is provided for the segmentation of the blood vessel.

3.4. Hybrid DR detection model

The feature extracted images are then given to the classification process. This paper proposes a hybrid based DL approach called the binocular Siamese based hybrid AlexNet-GoogleNet-SVM model. The binocular Siamese is the network-based model that can simultaneously manipulate the retinal fundus image and group their extracted features to make a probability prediction in each image component. Generally, the right and left eyes are scanned during the fundus examination at the clinical centers. It helps to identify retinal disease by discriminating the physiological and pathological conditions in a single eye. The unaffected eye plays a significant role in diagnosing the other eye. In simple words, if the right eye is severely affected due to DR, it can be proved that the person suffers from diabetes mellitus for a longer period, and the other eye is likely to have minor symptoms. It can be easily predicted using a binocular Siamese based network model.

Fig. 2 shows the simple representation of the binocular Siamese-based network model. In the simple block diagram, two inputs are given to two network models, and all extracted features are combined to obtain the outcome.

The CNN-based network models such as AlexNet and GoogleNet have multiple applications in the medical field. The AlexNet and GoogleNet are pre-trained models for classifying images of DR (normal and abnormal). In the DR source image, convolution, integration, and pooling layers are given to every image. In the ordinary approach, the softmax layer is used for classification purposes. The proposed model uses SVM as the classification process to get highly accurate images. The hybrid AlexNet-GoogleNet-SVM model approach framework is shown in Fig. 3.

3.4.1. Convolution layer

The convolution layer is very good for perceiving the image's texture. Not all extracted images can be processed to train the process. To eliminate the unwanted featured images, the CNN layer is used. In the convolution layer, S filtering is evaluated to increase the size of the data as $H \times G \times L$ where, H denotes the width, G denotes the height and L indicates the depth of the source image. The outcome of height and width is expressed in the mathematical formulation below,

$$\text{width outcome} = \left(\frac{H - U_b + 2d}{K_b} \right) + 1 \quad (38)$$

$$\text{Height outcome} = \left(\frac{H - U_n + 2d}{K_n} \right) + 1 \quad (39)$$

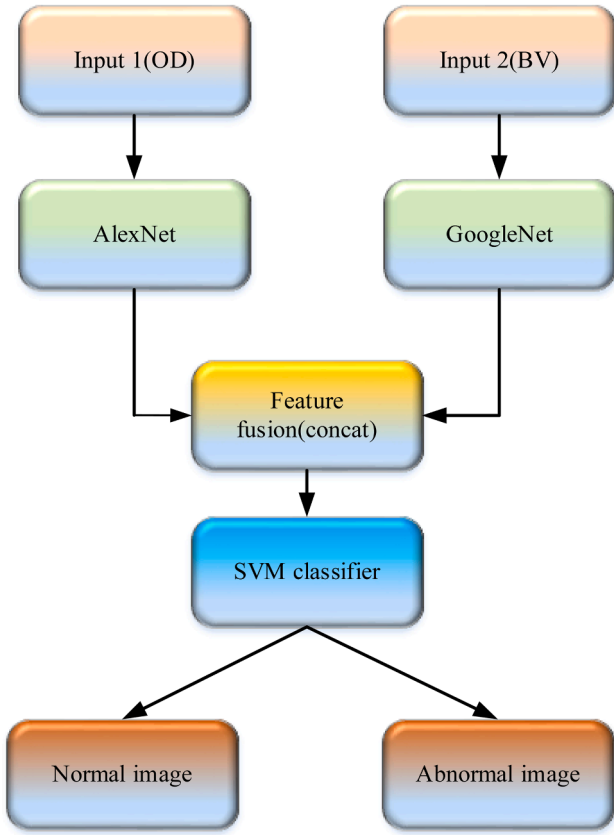


Fig. 2. Binocular Siamese network model.

Here, U_b, U_n, K_b, K_n, d denotes the filter height, filter width, stride width, stride height and padding, respectively.

3.4.2. Activation layer

The layer that comes after the convolution is the activation layer. This activation layer helps remove the linearity of the image. The activation layer is integrated with the rectifier (ReLU) function for efficient speed operation. The ReLU functional unit is mathematically formulated as given below,

$$relu = g(y) = \begin{cases} 0, & y < 0 \\ y, & y \geq 0, g(y)' = \begin{cases} 0, & y < 0 \\ 1, & y \geq 0 \end{cases} \end{cases} \quad (40)$$

3.4.3. Pooling layer

The pooling layer is often used between the convolutional layer to reduce the shift size of the image. There are many pooling layers: average, max, and L2-norm. The result of the pooling layer is mathematically formulated as follows,

$$DM = \left(\frac{SM + 2d - g}{k} \right) + 1 \quad (41)$$

Here, SM, DM and G denotes the source matrix, the decoded matrix and the stride, respectively.

3.4.4. Normalization

The output from convolution and fully convoluted layers are integrated to increase the computation time of the network. The integrated equation is mathematically expressed as,

$$H_j = \frac{J_j - \beta_1}{\sqrt{\alpha_1^2 + E}} \quad (42)$$

$$\alpha_1 = \frac{1}{U} \sum_{j=1}^U (J_j - \beta_1)^2 \quad (43)$$

$$\beta_1 = \frac{1}{U} \sum_{j=1}^U J_j \quad (44)$$

Here, U, β_1, α_1 , and H_j denoted the source image, average SD, and upcoming values from the integrated process.

3.4.5. Inception modulation

Convolution and max-pooling layers are completely different in this size. The parallel execution is used to expand the size of the inception module as $1 \times 1, 3 \times 3, 5 \times 5$ filters. The optimization of the processing load is the important purpose of this module.

3.4.6. Concatenate layer

The pooling layer and average pooling layer of the GoogleNet and AlexNet are provided with the feature vector. These feature vectors are merged using concat layer. The feature vectors have different sizes, and a padding process is introduced to overcome this issue. Hence, joining the two feature vectors produces a single feature vector. The maximization process is involved and applied to both pre-trained network models to represent these vectors.

Classification based on the SVM model: The SVM is a statistical learning theory that helps to implement structural risk reduction and common ability as the outcome. This model also works efficiently with small data sets. In simple words, SVM can solve the high dimensional and incomplete data. It is various applications like speech recognition, emotion recognition, fault diagnosis Parkinson's disease detection, Alzheimer's disease detection etc. The classification process that takes place in the SVM help in building an optimal separating hyperplane that expands the boundary among binary neighbouring data points corresponding to the different classes.

In the dataset $\{g_j, h_j\}, j = 1, 2, \dots, n$, the mathematical formulation of the linearly separable two-class classification is denoted as,

$$g(u) = \langle s, u \rangle + c \quad (45)$$

Here, s is the weight vector and c is a bias. The optimization problem for overcoming the optimal separable hyperplane is manipulated as follows,

$$reduction_{s,c,E} = \frac{1}{2} \langle s, s \rangle + D \sum_{j=1}^n E_j \quad (46)$$

$$Emphasize\ to : x_j(\langle s, n_j \rangle + c) + E_j - 1 \geq 0, E_j \geq 0 \quad (47)$$

Here, D the penalty parameter dominates the trade-off between the multiple decision function and the misclassified training values. E denotes the positive slack variable. The existing optimization model can be overcome by creating Lagrange multipliers and using the karush-kuhntucker theorem of optimization that shows the resultant as,

$$s = \sum_{j=1}^n \mu_j x_j n_j \quad (48)$$

Here, n_j denotes the positive Lagrange multipliers, μ_j denotes the support vectors that describe the decision boundary. The n_j parameter corresponds to 0, μ_j considered irrelevant. The μ_j^* is the optimal solution. The hyperparameters l^* and c^* are considered the optimal outcome. The linear separable binary classification problem in SVM for the differential function is denoted below,

$$h(n) = \text{sign} \left(\sum_{j=1}^n x_j \mu_j^* \langle n_j, n \rangle + c^* \right) \quad (49)$$

Considering the high-dimensional feature in the kernel function that

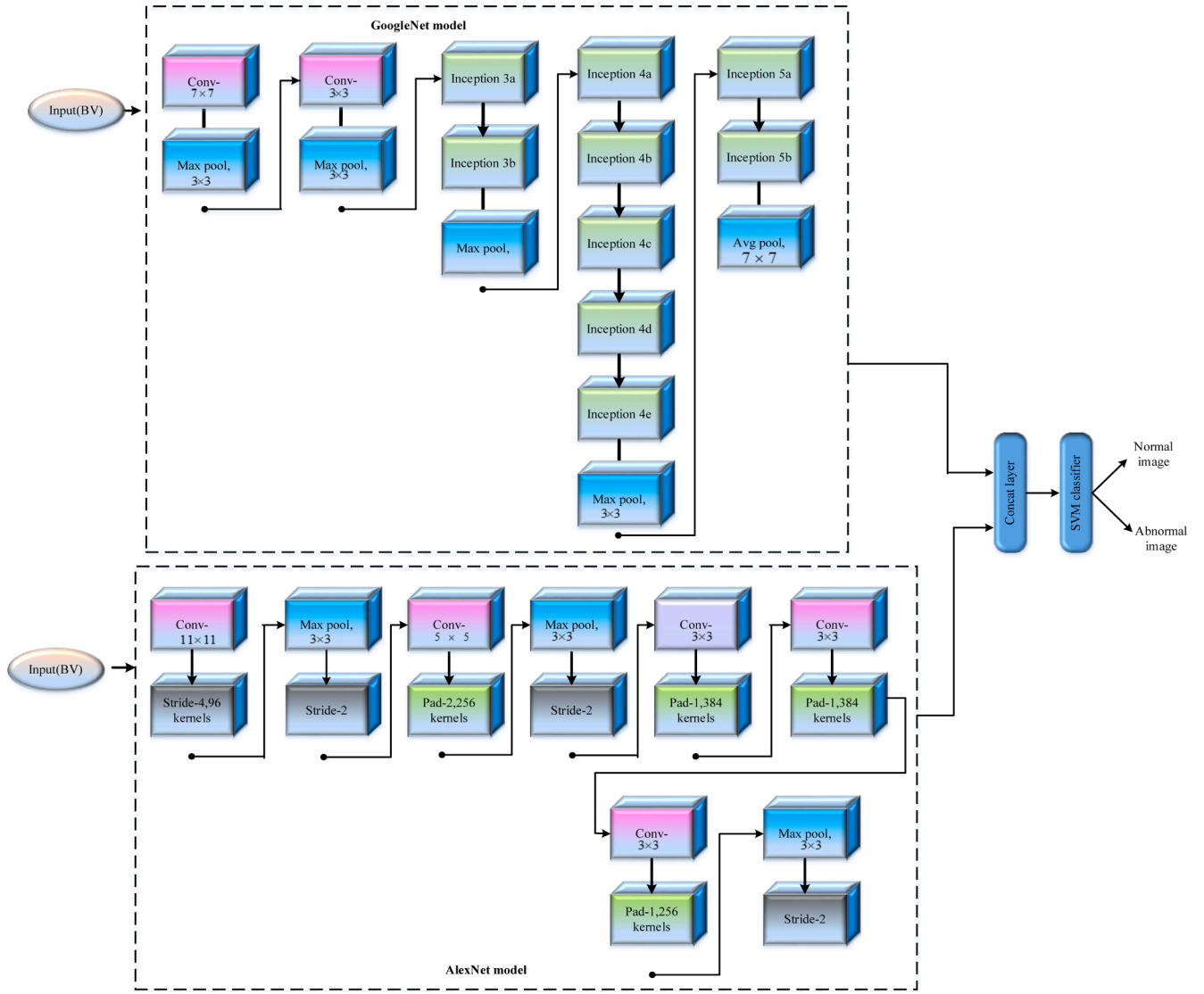


Fig. 3. Schematic diagram of Hybrid binocular Siamese AlexNet-GoogleNet-SVM model.

trains the SVM classifier based on non-linear maps. The kernel function is denoted by, μ and here the linear separation is highly possible. The mercer kernel function M denoted as $M\langle n, n_j \rangle = \langle \eta(n) * \eta(n_j) \rangle$ is manipulated based on scalar product $\langle \eta(n) * \eta(n_j) \rangle$. The mathematical formulation for the SVM classifier is denoted as,

$$h(n) = \text{sign} \left(\sum_{j=1}^m x_j \mu_j^* M\langle n, n_j \rangle + c^* \right) \quad (50)$$

Based on the understanding, the polynomial kernel with base 2 is enhanced for the SVM. Consecutively, the global kernel is used in data points that affect the kernel parameters due to far away distance. The general polynomial for the kernel function is formulated in mathematical form as,

$$M(n, n_j) = ((n_j \cdot n) + 1)^b \quad (51)$$

Here, b denotes the utilized polynomials. Fig. 3 demonstrates the schematic of the hybrid binocular Siamese AlexNet-GoogleNet-SVM classifier.

4. Result and discussion

This section presents the results of the developed approach to show

the efficient operation in contrast to other existing approaches. The proposed work is implemented using the PYTHON tool. This work has three phases: pre-processing, feature extraction, and classification. In the pre-processing phase, the images with noise are heavily removed, and the pre-processed image outcomes are shown in Fig. 4. The pre-processed images are then given to extract the features. The extracted features are shown in Fig. 5. The segmented images, such as optical disc and blood vessels, are present in Fig. 6. Finally, the segmented and the extracted images are fed to the classification stage, and the classified outcome is shown in Fig. 7. The work has been emphasized via two DR data sets such as DIARETDB0 and DIARETDB1, with the retinal fundus image. The dataset used in the proposed method is briefly explained below.

4.1. Dataset description

The proposed method uses DIARETDB0 and DIARETDB1 datasets consisting of retinal fundus images. The DIARETDB0 dataset consists of 130 RGB fundus images. Of this, 20 are normal, and the remaining 110 are symptoms with DR. These images are fetched from Kuo-Pio University Hospital of about 130 images. From the dataset images, 80 % are used as training data and 20 % as testing data. These 130 images are separated into 4 classification, which includes (i) Red tiny dots,

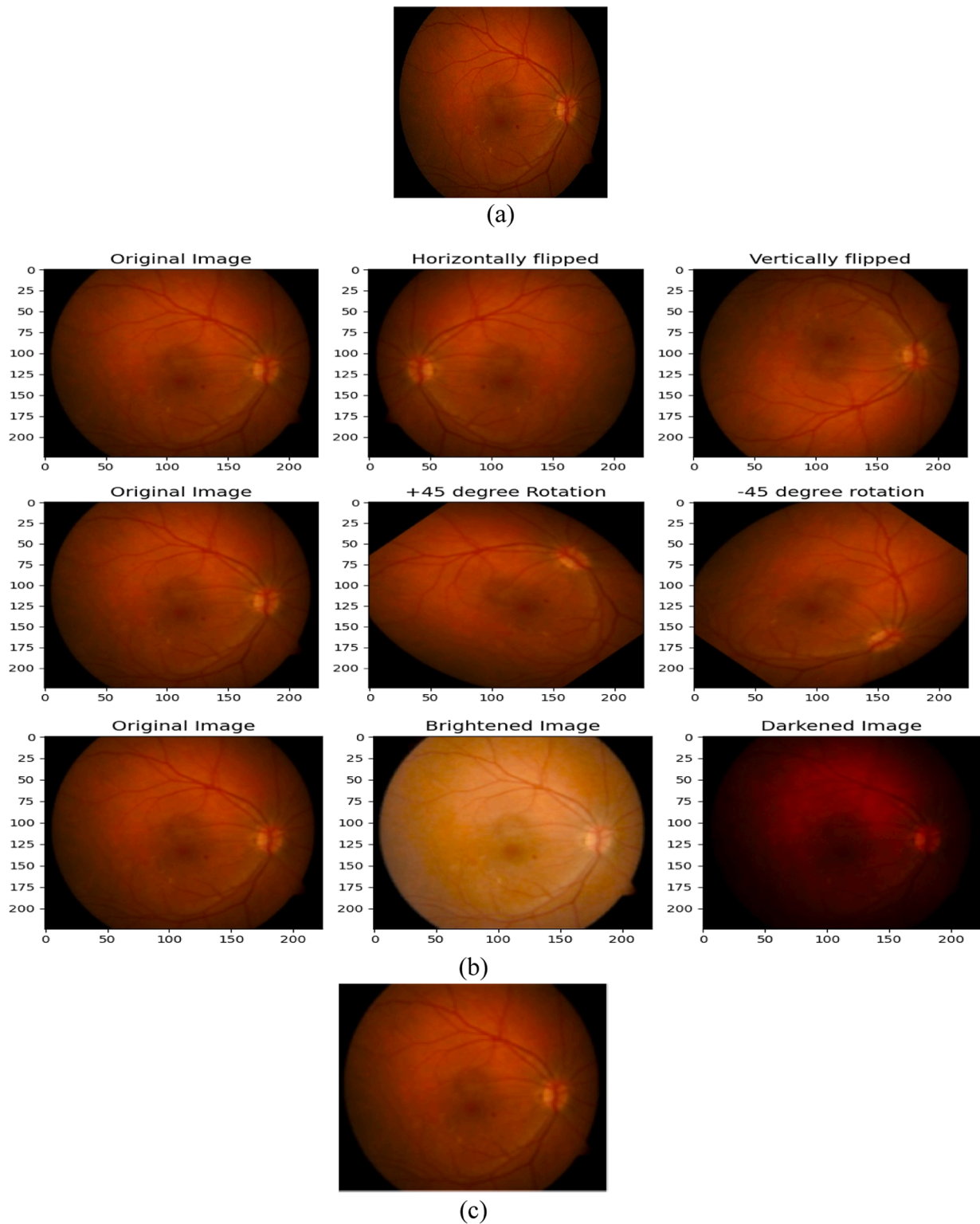


Fig. 4. Pre-processed image results, (a) Input image, (b) Augmented image, (c) Filtered image.

hemorrhages, temporal diffusion, (ii) Red tiny dots, hemorrhages, temporal diffusion, soft diffusion, (iii) Red tiny dots, hemorrhages, soft diffusion, neovascularization, (iv) Normal. The images are encapsulated in a 50-degree field view with the settings in the camera based on illumination, strength, shutter, slotting etc. The image consists of the fovea, optical disc (OD), macula, and blood vessel (BV) used to extract features.

The images also consist of an unknown amount of noise and optical aberrations with an unknown accuracy of photometric information. The

variance of the different DR images is maximal. Practically, the available data can able to calculate the general performance of the diagnosis methods. This dataset is also called calibration level 0 fundus image. The DIARETDB1 dataset consists of 89 RGB fundus images, of which 84 images contain minor DR symptoms. The remaining 5 images are normal with no symptoms of DR. This dataset is also known as measurement 1 fundus image. The images are encapsulated using multiple cameras with high noise and optical destruction, and it is finally known as

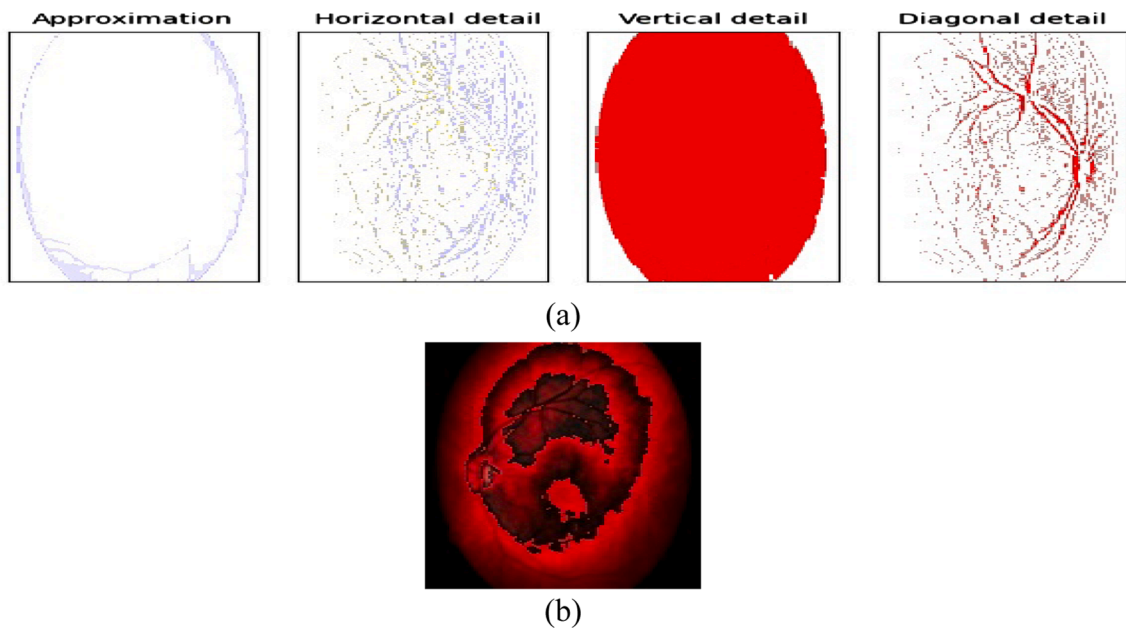


Fig. 5. Extracted band image and enhancement of LL band image, (a) Band image, (b) Enhanced image using COA technique.



Fig. 6. Input image with segmented BV and OD, (a) Input image, (b) Segmented BV, (c) segmented OD.

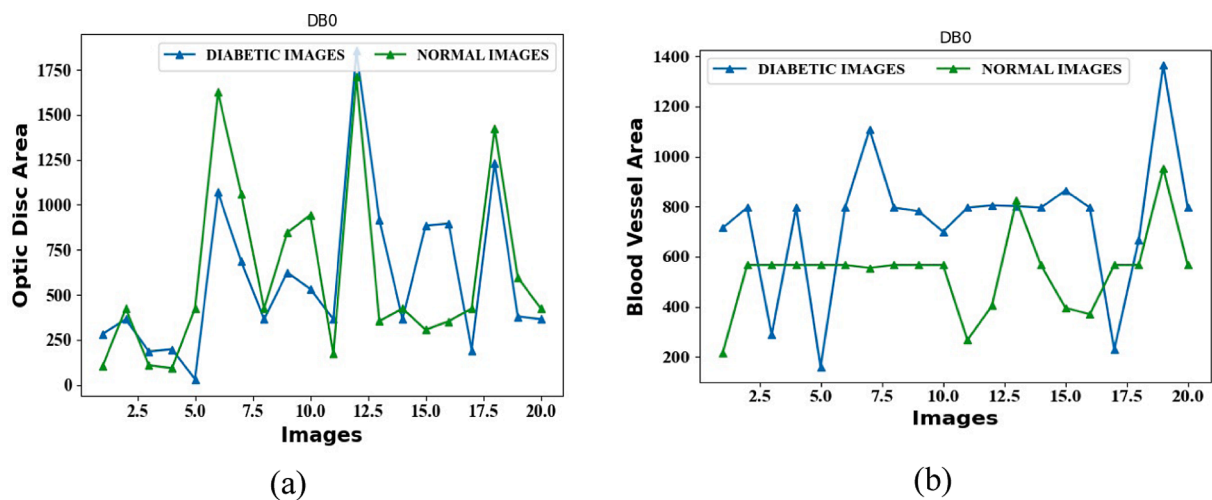


Fig. 7. Segmented Area of optical disc and blood vessel in DB0 dataset.

measurement level 0 fundus image. In this dataset, no calibration is done hence, the physical measurement values cannot be recovered. In the training set, 18 images are hard exudates, 6 soft exudates, 19 micro aneurysms and 21 hemorrhages. In the testing set, 20 images are hard exudates, 9 soft exudates, 20 micro aneurysms and 18 hemorrhages.

Table 1 tabulates the details of hyperparameters of the proposed method. Table 2 illustrates the Device configuration for the proposed model.

Table 1
Details of Hyperparameters.

S. No	Parameters	Values
1	Input size	224 × 224 × 3
2	Activation	ReLU layer
3	Optimizer	Adam
4	Batch size	32
5	Epochs	50
6	Learning rate	0.01
7	Hidden layers	5
8	Dropout	0.4

Table 2
Device configuration.

S. No	Parameters	Configuration
1	Device name	SSM107.smg.local
2	Processor	Intel (R) Core (TM) i5-470S CPU @ 2.90 GHz 2.90 GHz
3	Installed RAM	8.00 GB (7.88 GB unstable)
4	System type	64-bit operating system
5	Pen and touch	No pen or touch input is available for this display

4.2. Performance metrics

The performance metrics are accuracy, precision, recall, F1 score, specificity, FPR, FNR, NPV, FDR and MCC. The mathematical model is given as follows:

$$\text{Accuracy or } A_y = \frac{Tp + Tn}{Tp + Tn + Fp + Fn} \quad (52)$$

$$\text{Precision or } P_n = \frac{Tp}{Tp + Fp} \quad (53)$$

$$\text{Sensitivity} = \frac{TP}{TP + FN} \quad (54)$$

$$F - \text{measure} = \frac{2 \times \text{Precision} \times \text{Sensitivity}(\text{Recall})}{\text{Precision} + \text{Sensitivity}} \quad (55)$$

$$\text{Specificity or } TNR = \frac{TN}{TN + FP} \quad (56)$$

$$FPR = \frac{Fp}{Fp + Tn} \quad (57)$$

$$FNR = \frac{Fn}{Fn + Tp} \quad (58)$$

$$NPV = \frac{TN}{TN + FN} \quad (59)$$

$$FDR = \frac{FP}{(FP + TP)} \quad (60)$$

$$MCC = \frac{TP * TN - FP * FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (61)$$

4.3. Analysis of DR model

The proposed work mainly aims to detect and classify the DR in diabetic Mellitus affected patients. The oncoming parts describe the analysis performance of the DR image.

4.3.1. Pre-processing and data augmentation:

The image is augmented to 45 and -45 degrees in the pre-processing stage. After augmentation, the image with noise is removed using a cross guided bilateral approach. The guided and input image is utilized to

calculate the weight of the image. The bilateral filtering helps to eliminate the image illumination and examine the weak, noisy signal. Fig. 4 shows the pre-processed image results. Figure (a), (b) and (c) indicates the input image, horizontal and vertically flipped augmented image under + 45 and -45 degree and de-noised image, respectively.

4.3.2. Feature extraction stage

The pre-processed image is given to extract the features. In the proposed work, a wavelet based chimp optimization algorithm is used. This approach extracts Shannon, Renyi, Kapur entropy, and texture features. The DWT approach obtains the HH, HL, LH and LL bands. As a result, all other bands, except the LL band, achieve better results. The chimp optimization algorithm is brought into the LL-band to improve output to get the efficient LL-band. Fig. 5 below shows the different bands like LL, HL, LH, and HH and their corresponding enhanced image (LL band). In figure (a), the first figure shows the LL band, the second figure shows the LH band, the third figure shows the HL band, and the last figure shows the HH band. Figure (b) represents the enhanced retinal image.

4.3.3. Segmentation:

In this stage, the optical disc (OD) and blood vessel (BV) are segmented using the OCWSM technique. The optical disc, also known as the optical nerve head, is the point of outcome in the ganglion cell axons that leaves the eye. The blood vessel is the tiny tissue enclosing the eye's white part. The DR in patients may affect the retina of the eye. The segmented OD and BV are shown in Fig. 6. Figure (a), (b) and (c) represents the input image, segmented BV and segmented OD, respectively.

The segmented area curve for the optical disc and the blood vessel in the normal image is shown in Fig. 7. By varying the 20 images, the area of the optical disc is illustrated in Fig. 7a for the DIARETDB0 dataset. The comparison given below is based on diabetic and normal images. In addition, the 20 images are again compared with the blood vessel area based on the normal and abnormal images. The comparison of the area curve for BV is illustrated in Fig. 7b. The segmented area curve for OD and BV for DIARETDB1 is illustrated in Fig. 8. By varying the 20 images, the area of the OD is illustrated in Fig. 8a for the normal image. The comparison provided below shows the variation between normal and retinopathy images. However, the 20 images are again compared to the BV area based on normal and retinopathy images. The discrimination among the BV area curve is illustrated in Fig. 8b.

4.4. Statistical analysis of DR model

Table 3 mentioned below represents the statistical analysis for the DB0 dataset. The mean and SD of wavelet features, law, LBP, OD, BV, exudates, Shannon, Renyi, and Kapur, are tabulated in table 3. The statistical values obtained for normal and diabetes affected images are shown in the table. Table 4 mentioned the statistical analysis for mean and SD for the DB1 dataset. Various features like wavelet features, law, LBP, OD, BV, exudates, Shannon, Renyi and Kapur are extracted, and results obtained for normal and diabetic images are provided below.

4.5. Classification of DR model

The proposed method is a hybrid based binocular Siamese with an AlexNet-GoogleNet-SVM model to classify DR images. The upcoming section briefly explains the analysis performance of the proposed model.

4.5.1. Analysis of hybrid based binocular Siamese with AlexNet-GoogleNet-SVM model:

The proposed approach is manipulated with the DB0 and DB1 datasets to examine the method's performance based on the input images. The AlexNet-GoogleNet-SVM model is used to overcome the accuracy value, and the complexity of the dataset is also reduced. The accuracy and loss curve performance based on training and testing are

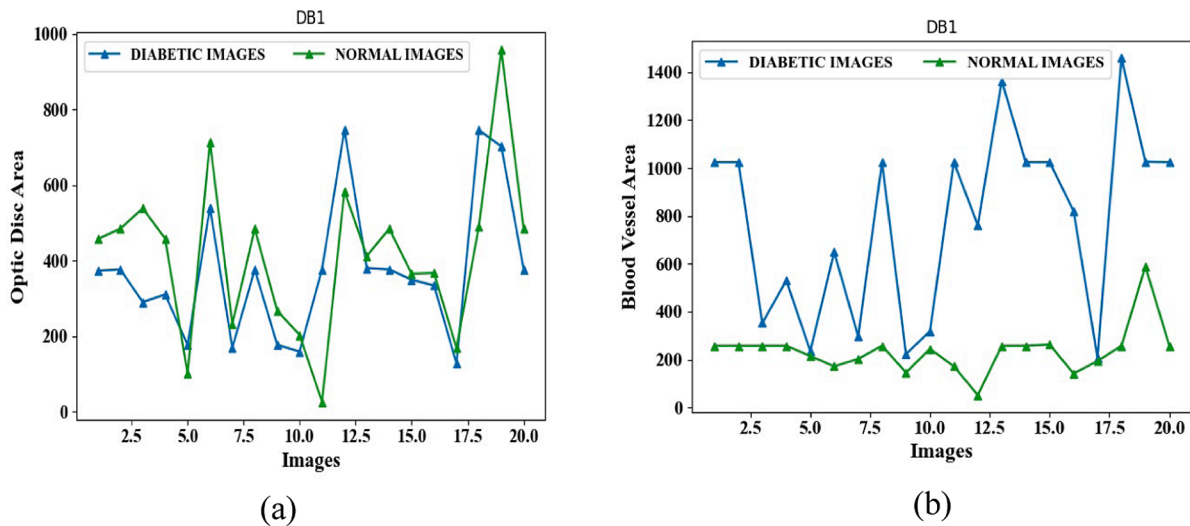


Fig. 8. Segmented Area of OD and BV in DB1 dataset.

Table 3
Statistical analysis of the DB0 dataset.

Features	Mean value		Standard deviation	
	Normal	Diabetic	Normal	Diabetic
Wavelet features	0.1646	0.1495	0.3294	0.2972
Law	43.6188	26.9582	4.7817	1.5375
lbp	0.0661	0.0429	0.1131	0.1240
Optic disc	6.5323 e + 04	6.5105 e + 04	131.4224	44.5145
Blood vessel	6.1562 e + 04	6.1731 e + 04	269.5542	171.8053
Exudates area	254.9757	254.9684	0.0063	0.0279
Shannon	3.2579	3.2711	1.3927	1.4297
Renyi	5.1610	5.0707	0.2665	0.6465
Kapur	3.5342	3.5389	0	0

Table 4
Statistical analysis of DB1 dataset.

Features	Mean value		Standard deviation	
	Normal	Diabetic	Normal	Diabetic
Wavelet features	0.0873	0.0955	0.1730	0.1926
Law	29.6218	31.0825	1.4910	3.6806
lbp	0.0453	0.0474	0.1231	0.1223
Optic disc	6.4732 e + 04	6.5375 e + 04	46.3596	186.2471
Blood vessel	6.1861 e + 04	6.2462 e + 04	81.6063	168.4366
Exudates area	0	0	0	0
Shannon	3.2504	3.2062	1.3865	1.4546
Renyi	5.1073	5.0249	0.5037	0.7465
Kapur	3.5063	3.5389	0	0

provided below. The extracted frames are given as input to the proposed model for the proposed model. In addition, the accuracy and loss curves are discriminated against by the training, testing and validation data. The accuracy and loss curve is distinguished for the experiment with 80 % training data, 10 % testing data and 10 % validation data. The graphical representation of the accuracy and loss curve for the DB0 dataset is displayed below.

Fig. 9 illustrates the accuracy and loss curve for training and testing data in the DB0 dataset. The comparison is made with training and test data in the accuracy curve, as shown in Fig. 9a. According to the graph, the proposed model achieved a low or high balanced score on both graphs. When the number of iterations is high, the accuracy is also high. The proposed network model is trained for 100 iterations, and the overall accuracy obtained for the DB0 dataset is 96 %. The graph above

shows that the model is highly stable for increased iterations for training and testing the data.

The comparison is made with training and testing data in the loss curve, as shown in Fig. 9b. According to the graph, the loss percentage for the developed model is highly decreased, and if the number of iterations is high, the outcome will be very low. From this graph, it can be seen that the proposed model is highly stable with increased iterations. In the experimental scenario, the loss value obtained is 0.03 for 100 iterations compared to other existing approaches. Thus, the proposed model attains a low error since the training data used is highly optimal because of the effective developed model.

Fig. 10 demonstrates the accuracy and loss curve for training and testing data in the DB1 dataset. The comparison is made with training and testing data in the accuracy curve, as shown in Fig. 10a. According to the graph, the proposed model obtained a low or high equalized outcome in both graphs. When the number of iterations is high, the accuracy is also high. The proposed network model is trained for 100 iterations, and the overall accuracy obtained for the DB1 dataset is 97 %. The graph above shows that the proposed model is highly stable for increased iterations for both training and testing the data.

The comparison is made with training and test data in the loss curve, as shown in Fig. 10b. According to the graph, the loss percentage for the developed model is highly decreased, and if the number of iterations is high, the outcome will be very low. From this graph, it can be seen that the proposed model is highly stable with increased iterations. In the experimental scenario, the loss value obtained is 0.02 for 100 iterations compared to other existing approaches. Thus, the proposed model attains low error as the training data used is highly optimal because of the effective developed model.

The outcome of the proposed model is determined using the decoded operating region under the curve (ROC). Fig. 11 indicates the ROC curve for the DB0 dataset. ROC is designated based on the true positive and false-positive rates. By varying the FPR, the TPR is increased consecutively. The developed model attains the area under the curve (AUC) of 95 % for detecting DR images for the DB0 dataset.

The outcome of the proposed model is determined using the decoded operating region under the curve (ROC). The ROC curve is designated based on the true positive and false-positive rates. The ROC curve for the DB1 dataset is shown below in Fig. 12. Consecutively, by changing the FPR, TPR is highly increased. The developed model attains the area under the curve (AUC) of 98 % to detect DR images for the DB1 dataset.

4.5.2. Performance analysis of classification model:

The performance analysis of optic disc classification and blood vessel

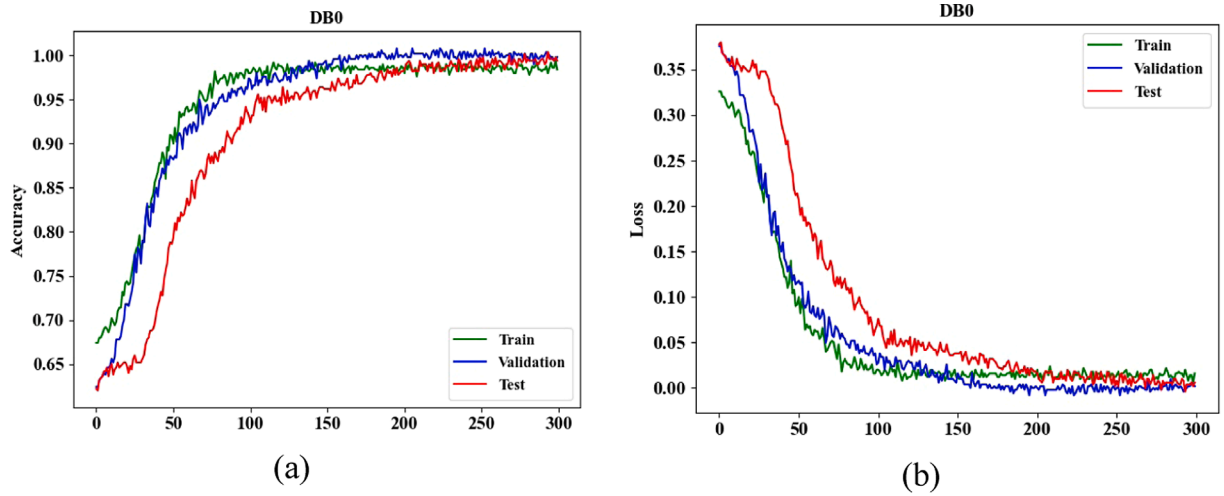


Fig. 9. Accuracy and loss curve for training and testing data in DB0 dataset (a) Accuracy curve (b) loss curve.

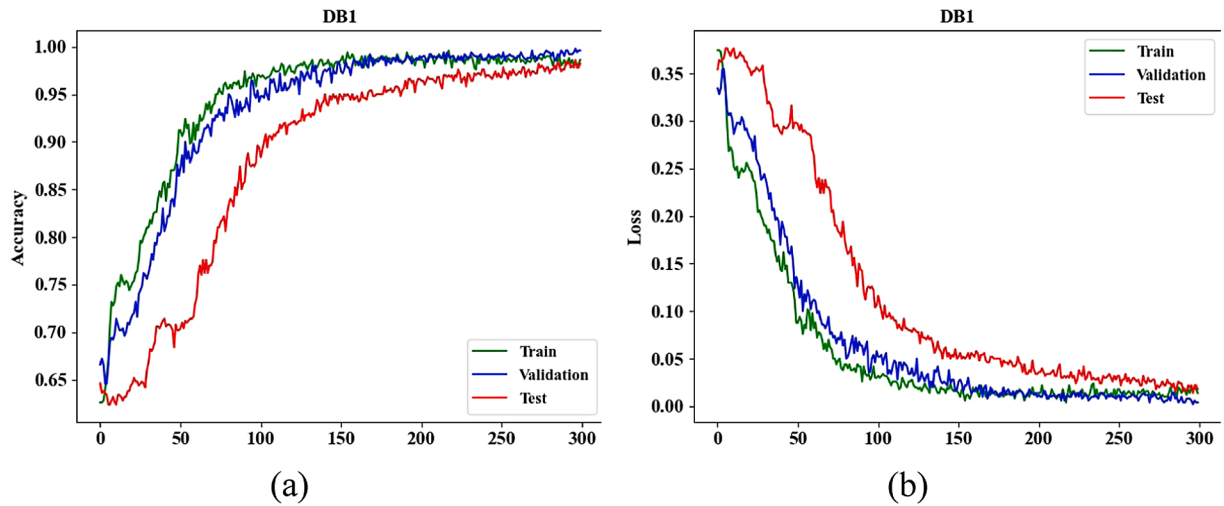


Fig. 10. Accuracy and loss curve for training and testing data in DB1 dataset (a) Accuracy curve (b) loss curve.

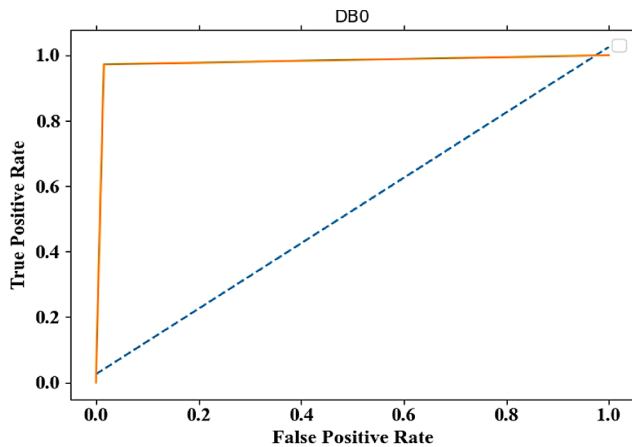


Fig. 11. ROC Curve for DB0 dataset.

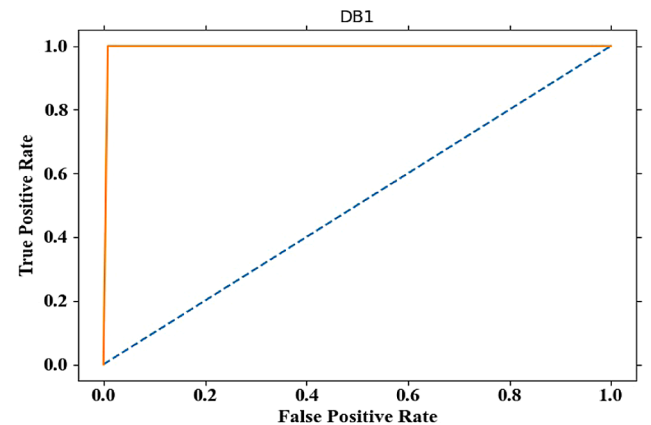


Fig. 12. ROC Curve for DB1 dataset.

classification is explained below:

(a) Performance analysis of optic disc for DB0: The accuracy, precision, and F1-score performance metrics are analyzed for the optic disc for the DB0 dataset and compared with various existing methods: DBN

[29], NN [29], DBN + NN [29], and TB-DA [29]. Fig. 13 shows the comparative analysis of optic disc on the DB0 dataset. Fig. 13(a) shows the accuracy of the proposed method. It is evaluated by varying the learning percentage as 35, 50, 65, 75, and 85. The achieved accuracy level of the proposed HBS-AGSVM is 93.25 %. If the learning percentage

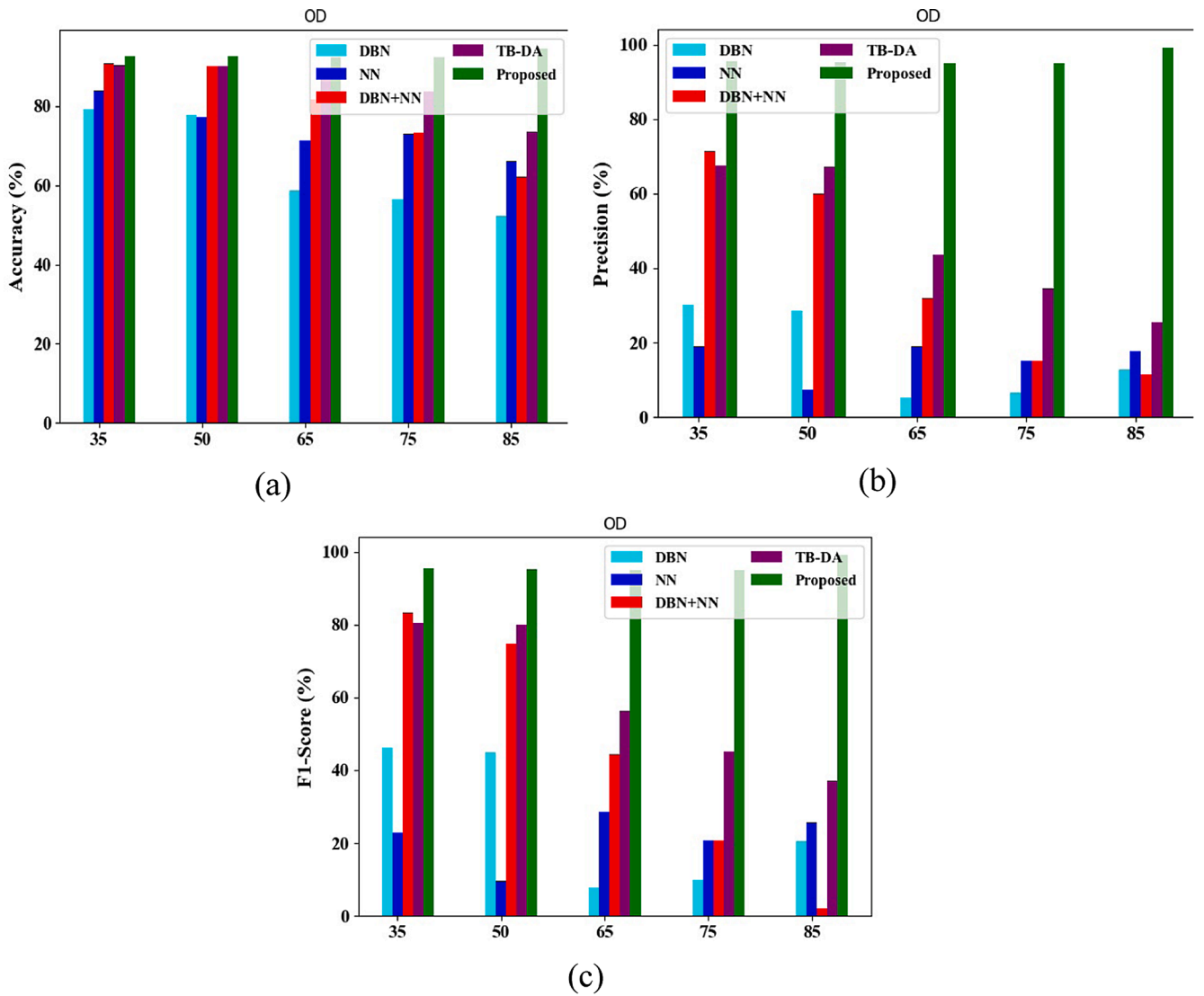


Fig. 13. Comparative analysis of optic disc in DB0 dataset (a) Accuracy (b) Precision (c) F1-score.

of 65 is considered, the proposed system is 40 % better than the DBN, 28 % better than the NN, 16 % better than the DBN + NN, and 10 % better than the TB-DA.

Fig. 13b illustrates the precision performance metrics comparison of various existing systems with the proposed model. Some of the existing methods which compared: are DBN [29], NN [29], DBN + NN [29], and TB-DA [29]. When the learning rate is 85, the attained accuracy values for previous systems and the proposed are 15 %, 18 %, 5 %, 28 % and 95.79 %, respectively. Fig. 13c shows the F1-score performance metrics of the proposed and existing optic disc comparison. The proposed methodology precision value is high, i.e. 96.77 %, compared with the existing approaches: DBN, NN, DBN + NN, TB-DA [29]. If the learning rate is 85, the precision value for the existing DBN is 20 %, the existing NN is 25 %, the previous DBN + NN is 5 %, and the previous TB-DA method is 38 %, correspondingly.

(b) Performance analysis of blood vessels for DB0: The proposed methodology of blood vessels of DB0 dataset performance analysis is evaluated based on accuracy, precision and recall. Fig. 14 indicates the comparative analysis of blood vessels for the DB0 dataset. Fig. 14 (a) clearly shows the accuracy level of the proposed HBS-AGSVM method. At the learning percentage of 85, the achieved accuracy rate of different existing methods such as DBN, NN, DBN + NN, TB-DA [29] and the proposed approach is 68 %, 75 %, 82 %, 88 % and 93.6 %, which means

that the proposed system attained a maximum accuracy level than the existing method.

The level of accuracy achieved by existing and proposed HBS-AGSVM is shown in Fig. 14b. The precision value of the introduced method HBS-AGSVM is 97.03 %, which is maximum than the various previous methods like DBN, NN, DBN + NN, and TB-DA [29]. The achieved precision rate of the proposed is 84 % enhanced than the DBN, 78 % enhanced than the NN, 60 % improved than the DBN + NN, and 40 % improved than the TB-DA when the learning rate is 85. The F1-score performance of the proposed and the different existing methods comparison is illustrated in Fig. 14c. The attained F1-score value for the proposed method is 97.55 %, much greater than the existing methods. By varying learning rates, the F1 measure of the existing and proposed system varies by considering the learning percentage 85. The proposed method is 85 % superior to the implemented DBN, 70 % superior to the existing NN, 42 % maximum than the previous DBN + NN, and 30 % superior to the implemented TB-DA approach. Table 5 illustrates the performance metrics comparison of normal and diabetes for the DB0 dataset.

(c) Performance analysis of optic disc for DB1 dataset: Fig. 15 indicates the performance comparison of optic disc for the DB1 dataset. The performance measures such as accuracy, precision and F1-score graphical representation for an optic disc of the DB1 dataset are

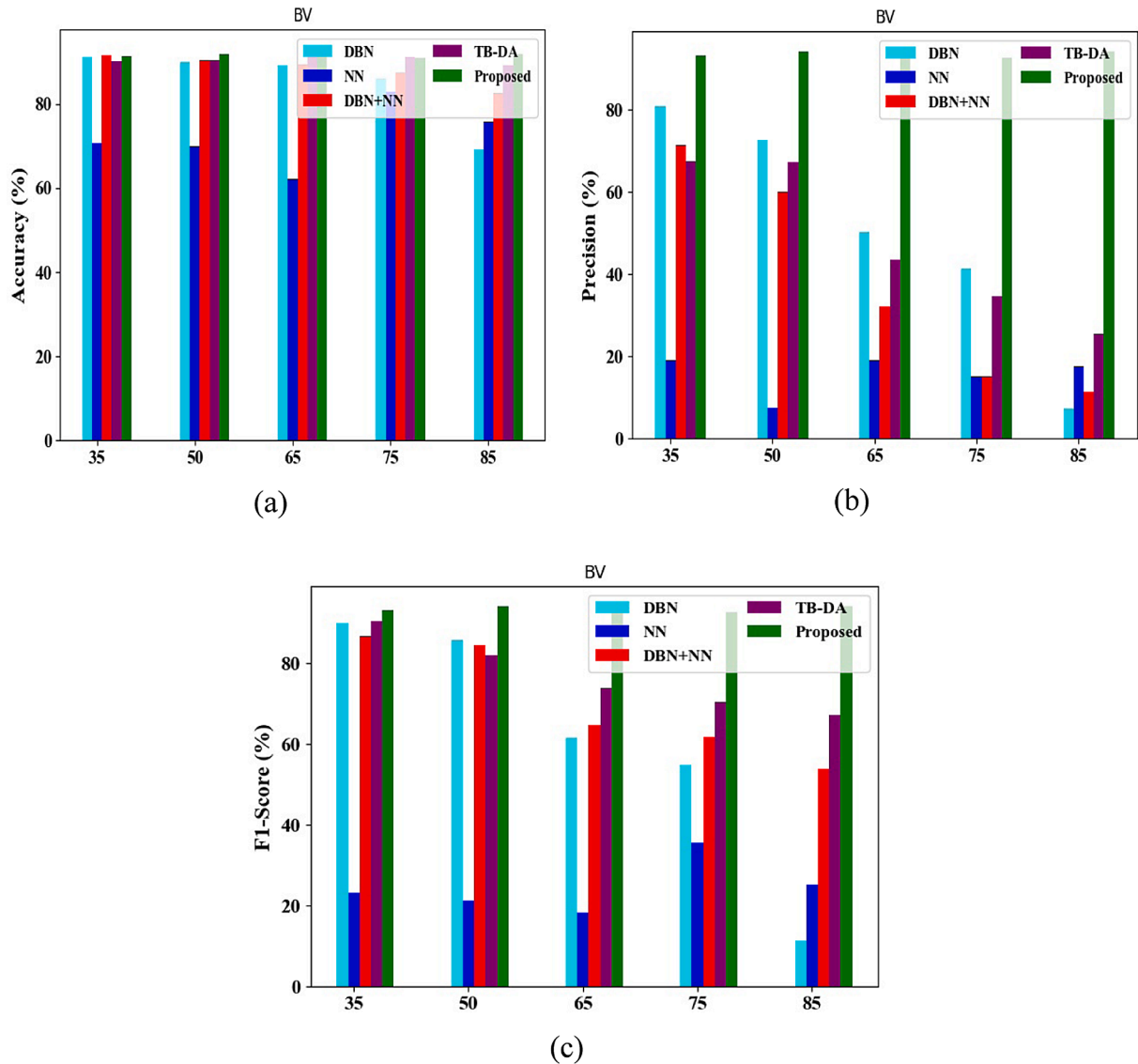


Fig. 14. Comparative analysis of blood vessel for DB0 dataset (a) Accuracy (b) Precision (c) F1-score.

Table 5

Performance metrics comparison of normal and diabetes for DB0 dataset.

DB0 dataset		
Performance metrics	Normal	Diabetes
Accuracy	93.25 %	93.68 %
Precision	95.79 %	97.03 %
F1-score	96.77 %	97.55 %
Recall	97.83 %	98 %
Specificity	97 %	98 %
FPR	0.021	0.01
FNR	0.02	0.03
NPV	0.04	0.02
FDR	0.044	0.03
MCC	0.93	0.95

mentioned in Fig. 15a, 15b and 15c, respectively. The proposed model is more effective because of the attained accuracy value, i.e. 94.18 %. The other existing methods compared to the proposed HBS-AGSVM are DBN, NN, DBN + NN, and TB-DA [29]. The accuracy values of existing and proposed methods in the learning percentage 85 are: DBN achieves 53 %, NN reached an accuracy result is 66 %, DBN + NN attained an accuracy level of 63 %, and the existing TB-DA achieves 75 %. These

accuracy values prove that the proposed model accomplishes a more accurate outcome for the optic disc in the DB1 dataset than the existing approaches. Fig. 15b mentions the proposed and existing optic disc's precision outcomes. If the learning rate percentage is 85 %, the obtained precision values of the proposed and the existing approaches DBN, NN, DBN + NN, and TB-DA [29] are 93.75 % and 15 %, 18 %, 18 %, 5 %, 25 %, accordingly. The proposed method attained precision values that are more maximum in the dataset of DB1 than in the previous models. Fig. 15c is the performance analysis of the F1 measure with the proposed and existing systems. The proposed system reaches 96.4 %, the maximum F1-score value compared with the implemented DBN, NN, DBN + NN, and TB-DA approaches. The HBS-AGSVM proposed system is 77 % superior to the existing DBN, 72 % improved to the NN, 94 % enhanced than the DBN + NN and 62 % maximum than the TB-DA when the learning rate is 85.

(d) **Performance analysis of blood vessels for DB1 dataset:** The comparison of the proposed method blood vessels for DB1 dataset performance metrics are accuracy, precision and the F1-score with the existing methods. Fig. 16 represents the performance comparison of blood vessels for the DB1 dataset. Fig. 16a shows the accurate comparison of the proposed and the existing methods. The proposed methodology reaches the accuracy level of 94.83 %, which is the maximum if

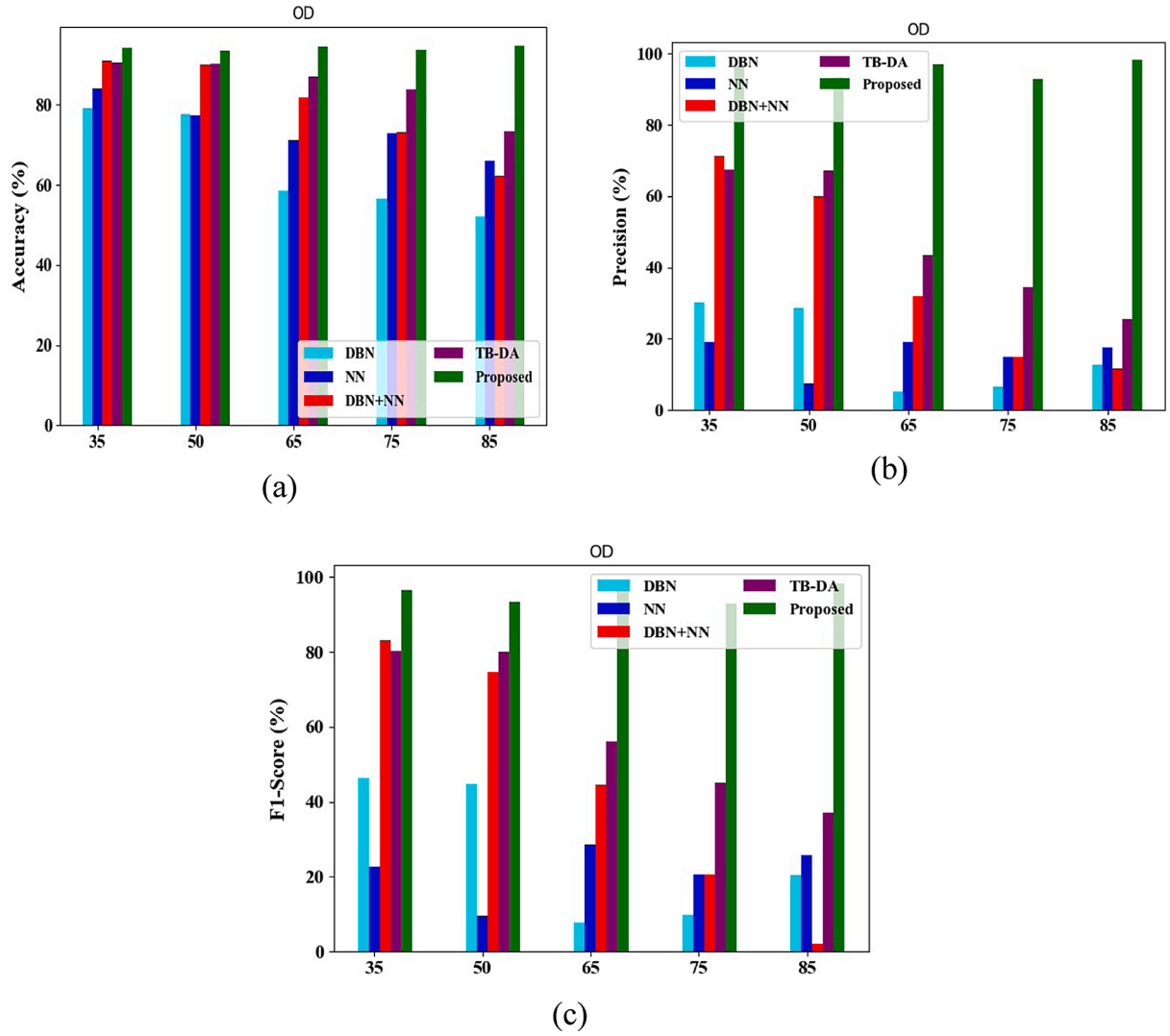


Fig. 15. Performance comparison of optic disc for DB1 dataset (a) Accuracy, (b) Precision, (c) F1-score.

compared to the implemented approaches. When the learning percentage is 85, the implemented DBN reaches 70 %, the existing NN reaches 75 %, the existing DBN + NN attains 82 %, and the implemented TB-DA accomplishes 88 % as the respective accuracy levels. It proves that the efficiency of the proposed method is maximum. Fig. 16b illustrates the precision analysis of various existing and proposed approaches. If the learning percentage is 85, the occurred precision value proposed is 91.9 %, which is the maximum, and some of the existing methods are DBN, NN, DBN + NN, and TB-DA [29], which attains minimum precision value such as 9 %, 20 %, 40 %, and 55 %.

Fig. 16c clearly shows the F1 measure of the proposed and the previously implemented models. The existing is compared with the proposed achieved F1-score values of 12 %, 25 %, 53 %, 69 % and 92.2 %, respectively, if the learning rate is 85. It shows that the proposed model efficiency is higher than the implemented models. Table 6 demonstrates the performance metrics comparison of normal and diabetes for the DB1 dataset.

Table 7 illustrates the comparative analysis of the proposed method on different methods and datasets. The table clearly shows that the proposed method obtains better outcomes than the existing techniques. The existing techniques attain poor outcomes due to high intra class variation, high illumination and background cluttering. But, the proposed method outperforms well by preventing high variation among the images. Table 8 illustrates the comparison of training time performance

with existing methods. The table shows that the proposed method consumes less training time than existing techniques. The existing techniques are highly affected due to high training time. This is because of the low capability of the network model. The existing raining model cannot train a huge number of datasets. But, the proposed hybrid DL model can compromise the overfitting issues, reducing the training time.

4.6. Discussion

The proposed work undergoes four stages: pre-processing, frame extraction, segmentation, and classification. The noises are filtered out in the pre-processing stage using the CGBF technique. The next stage is feature extraction, and in this stage, the feature is extracted using a hybrid wavelet-based chimp optimization algorithm is emphasized. This stage plays a dominant role in obtaining an efficient classification outcome. Moreover, in segmentation stage is implemented to segment the OD and BV from the extracted features. Finally, the DR image is classified using the binocular Siamese based AlexNet-GoogleNet-SVM model. The outcome of the proposed model is analyzed and compared with the previous approach, such as DBN, NN, DBN + NN and TB-DA models. But, the previous models achieve only low performance because of major disadvantages. The accurate outcome for classifying DR images such as normal and abnormal is very difficult in CNN approaches. In the existing approach, hyperparameters feature extraction

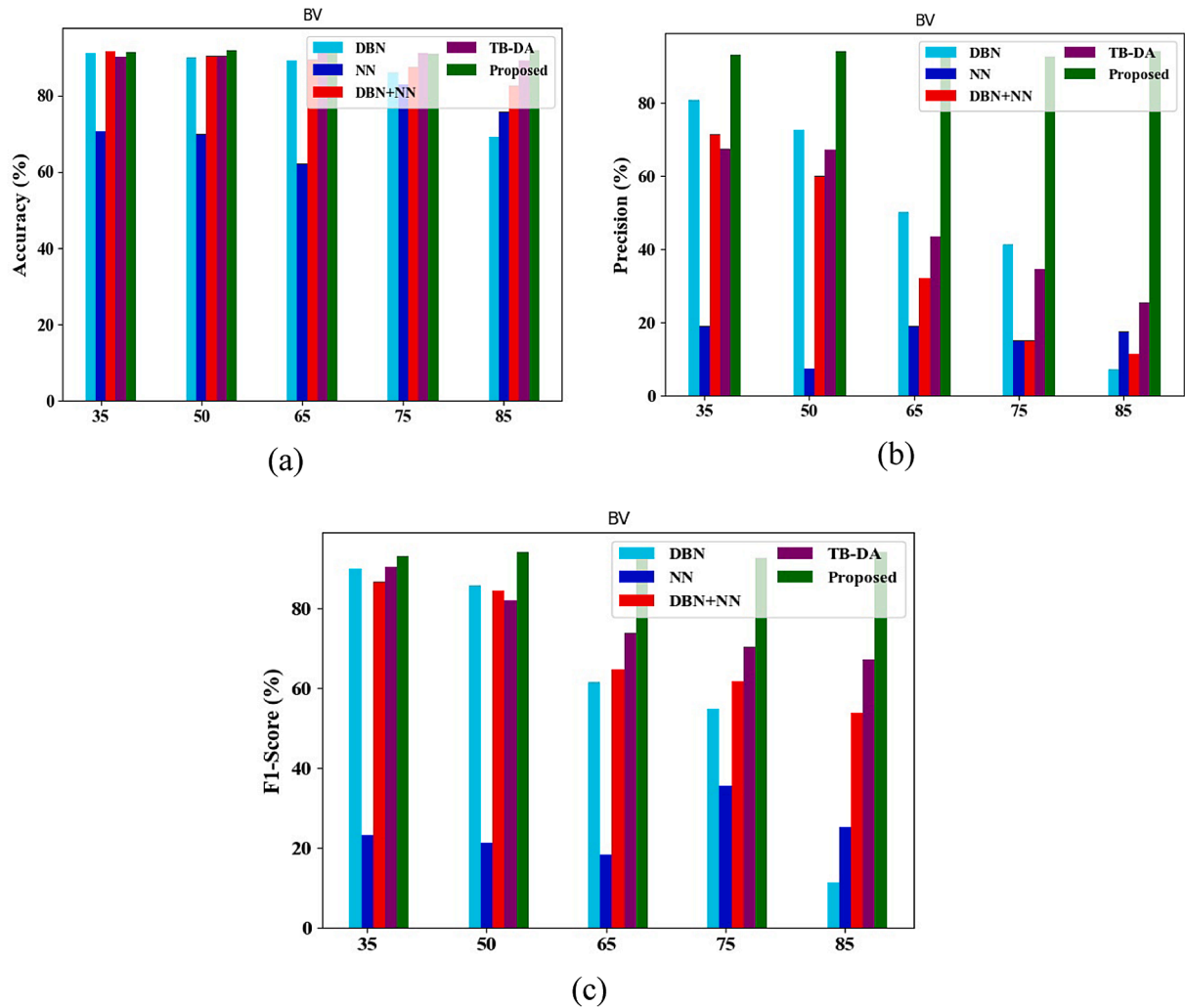


Fig. 16. Performance comparison of blood vessel for DB1 dataset (a) Accuracy (b) Precision (c) F1-score.

Table 6

Comparative analysis of normal and diabetes for DB1 dataset.

DB1 dataset		
Performance metrics	Normal	Diabetes
Accuracy	94.18 %	94.8 %
Precision	93.75 %	91.9 %
F1-score	96.45 %	92.2 %
Recall	99.5 %	93.26 %
Specificity	99 %	92 %
FPR	0.004	0.10
FNR	0.04	0.08
NPV	0.06	0.12
FDR	0.07	0.10
MCC	0.93	0.92

and tuning are better. But, overfitting issues may occur while using larger datasets. Hence, the accuracy of the CNN model is reduced to 83 %. The neural networks in the CNN layers are determined by integrating with the weight function. In the DBN model, the tuning of hyperparameters is not accurate and occurs in complex computation. Due to this drawback, this model attains low accuracy of about 85 %. The NN model was used to detect the DR images; however, the accuracy attained was about 93 %.

Some other models based on the traditional approach reach only accuracy of about 70 % due to poor detection of patients affected with DR disease. Proper selection of kernel function requires larger time, and

Table 7

Comparative analysis of proposed method on different methods and datasets.

Methods	Dataset	Recall (%)	Specificity (%)	Accuracy (%)	F-score (%)
Goel et al. [30]	IDRID	84	–	94.3	87.3
Bilal et al. [31]	DIARETDB0	95.99	96	93	94
Khan et al. [32]	kaggle dataset	55.6	91	85	59.6
Luo et al. [33]	MESSIDOR-2	80	79	77.75	–
Proposed	DB0	99.5	99	94	96.4
	DB1	93.2	92	94.83	92.23

hyperparameter tuning makes the system complex and time-consuming. The DBN + NN based approach is the existing approach introduced to detect the presence of DR affected images. But, this model attains a low accuracy of only 75 %. Due to a high mismatch in the outcome, this model attains poor accuracy. The adaptive threshold approach is used in TB-DA models to detect and classify the DR outcome. However, this model is highly suffered due to major drawbacks such as a high error rate and taking high time to tune the hyperparameters. The proposed model is executed with metrics like accuracy, precision, recall,

Table 8

Comparison of training time performance with existing methods.

Methods	Average training time (minutes)
QEIRV-2 [34]	44
KNN + random forest [35]	76.6
DenseNet-201 + ANN [36]	121
AlexNet + SVM [37]	537
GoogleNet [38]	40
Proposed	26.32

specificity, F1-score, FNR, NPV, FDR, and MCC by discriminating with the technique above. The developed model is a hybrid binocular Siamese-based AlexNet-GoogleNet-SVM model for efficiently classifying DR images. The weight function and tuning of hyperparameters are efficiently done using the developed model. Additionally, statistical measures are also analyzed based on mean and SD to investigate the efficiency of the proposed model.

In addition, performance is evaluated by changing the training percentage to 80 %. By altering the training percentage, the outcome based on accuracy, precision, recall, specificity, F1-score, FNR, NPV, FDR and MCC are determined and distinguished from previous approaches like DBN, NN, DBN + NN and TB-DA [29] models. Moreover, the softmax layer is highly affected due to multiple drawbacks because of classification problems. But, the SVM classifier used here obtains high efficient classification output. Because of this excellent performance, the proposed model attains high accurate outcomes.

5. Conclusion

This paper introduces a neural network model for diagnosing DR. The DR is recognized under three stages: Pre-processing, feature extraction, segmentation and classification. The pre-processing step eliminates noise using the cross-guided bilateral filter (CGBF). In addition, a wavelet-based chimp optimization method is used to extract the features from the retinal fundus image in the feature extraction phase. Some extracted features are Kapur entropy and Shannon entropy texture features, Renyis entropy, etc. After that, in the segmentation section, the optic disc and blood vessels get segmented with the help of the open-closed watershed management approach. Finally, the Binocular Siamese-like hybrid AlexNet – GoogleNet - SVM model is introduced to recognize the normal and diabetes images in the classification step. In addition, the proposed methodology undergoes training and testing along with loss and accuracy curves. The proposed method performance analysis is proved through the performance measures accuracy 94 %, precision 93.75 %, F1-score 96.4 %, recall 99.5 %, specificity 99 %, FPR 0.004, FNR 0.04, NPV 0.0625, FDR 0.07 and MCC 0.93 for DB0 dataset. Also, the results obtained from DB1 dataset are accuracy 94.83 %, precision 91.90 %, F1-score 92.23 %, recall 93.2 %, specificity 92 %, FPR 0.10, FNR 0.08, NPV 0.126, FDR 0.10, MCC 0.92. Furthermore, the statistical performance for the normal and diabetic images is also analyzed based on mean and SD. The proposed method can detect other abnormalities like Microaneurysms and hemorrhages from the retinal fundus image. Despite this, the proposed technique cannot apply to classifying mild and severe PDR and NPDR diseases from the fundus image. In future, the researchers must focus on developing the DL techniques for multi-class classification in DR detection. In addition, this technique cannot detect the cotton wool and white spots in the retinal part. In future, this work can be extended by applying enhanced AI based techniques on different datasets to detect the hard and soft exudates from the retina.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Data availability

The authors do not have permission to share data.

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